

Interpretability:

Ability to understand the decision-making process of an AI model. An interpretable model is transparent in its operation and provides information about the relationships between inputs and outputs. An interpretable algorithm can be explained clearly and understandably by a human being. Interpretability is therefore important to ensure that users can understand and trust artificial intelligence models.

Explainability:

Ability to explain the decision-making process of an AI model in terms understandable to the end user. An explainable model provides a clear and intuitive explanation of the decisions made, enabling users to understand why the model produced a particular result. In other words, explainability focuses on why an algorithm made a specific decision and how that decision can be justified.

Interpretability vs Explainability

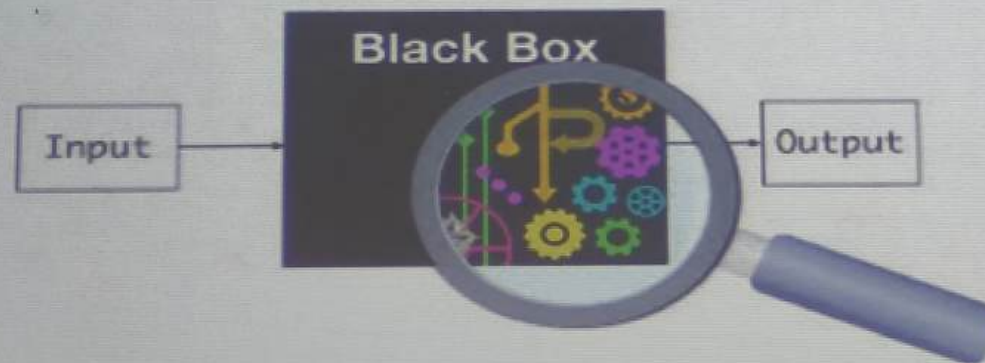
	Interpretability	Explainability
Definition	Interpretability refers to the degree to which a human can understand the cause of a decision made by a model. It involves making the model's workings clear enough so that users can comprehend how inputs are being processed to produce outputs.	Explainability goes a step further to not only make the model understandable but also to provide reasons for its decisions in a way that can be easily communicated to stakeholders. It's about providing insights into the model's behavior, particularly focusing on why a model made a particular decision.
Focus	Focuses on understanding the model itself.	Understanding and communicating the model's decisions.
Question	"How does this model work?"	"Why did the model make this prediction?"
Scope	Direct understanding of the model's inner logic	Post-hoc methods used to explain black-box behavior

Interpretability vs Explainability

	Interpretability	Explainability
Examples	Inherent- Linear Regression: The relationship between inputs and the output is a straight line, which is easy to understand and visualize. Decision Trees: The decision process is a series of if-then-else statements, which can be visualized and followed easily.	Post-hoc- LIME (Local Interpretable Model-agnostic Explanations): It provides local approximations of the model's behavior to explain individual predictions. SHAP (SHapley Additive exPlanations): It uses game theory to provide a consistent and fair attribution of feature importance.
Applicability	Applies to transparent models (e.g., Decision Trees, Linear Regression)	Used for complex models like Neural Networks, XGBoost
Techniques	Often involves inherently interpretable models or simplified versions of complex models.	Involves tools and techniques that can be applied to any model, providing explanations after the fact.
Target Audience	Primarily aimed at data scientists and model developers.	Aimed at a broader audience, including non-technical stakeholders who need to understand the model's decisions.

Black Box Model

- A Black Box Model is a system that does not reveal its internal mechanisms.
- In machine learning, “black box” describes models that cannot be understood by looking at their parameters (e.g. a neural network).
- The opposite of a black box is sometimes referred to as White Box.
- Essentially, users can see the input and output of the model but cannot interpret how the model arrived at its prediction.



Interpretability Methods-Techniques for Explaining Black-Box Models

- Partial Dependence Plots (PDP)
- Surrogate Models
- Permutation Feature Importance
- Grad-CAM (Gradient-weighted Class Activation Mapping)
- Saliency Maps
- LIME (Local Interpretable Model-agnostic Explanations)
- SHAP (SHapley Additive exPlanations)

Regression

- Regression is a supervised learning technique used to predict continuous (numerical) output values based on one or more input features.
- The goal is to learn a function that maps inputs (features) to a real-valued output.
- Ex. House Price Prediction, Weather Prediction, Stock Market Prediction, Cricket Score Prediction

Linear Regression

- Linear regression is a statistical method used to model the relationship between a dependent variable and one or more independent variables by fitting a linear equation to the observed data.
- In simpler terms, it aims to find the "best-fit" straight line through a set of data points, allowing for prediction of unknown values based on known ones.

$$y = b + w_1x_1 + w_2x_2 + \dots + w_nx_n + \epsilon$$

where,

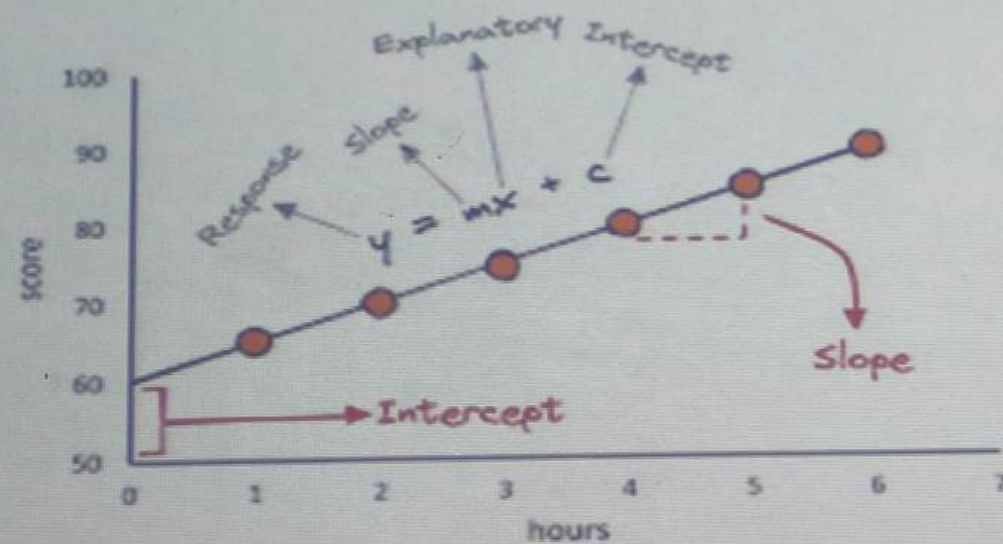
y is the dependent variable

b is the y -intercept

$w_1, w_2, \dots, w_n$ are the coefficients (slopes) for each independent variable

ϵ is the error term, representing the difference between the predicted and actual values

Linear Regression



Applying Interpretability to Linear Regression

- Linear regression models are inherently interpretable due to their linear nature, allowing for straightforward analysis of how independent variables influence the dependent variable.
- Coefficients- Show how much each feature contributes to the outcome
- Sign (+/-)- Shows whether the feature increases or decreases output
- Transparency- Everything is visible, explainable, and auditable

Ex. Predicting house price based on size (Area) and Age:

$$\text{Price} = 50,000 + 3000 \times \text{Area (sqft)} - 2000 \times \text{Age (years)}$$

Interpretation:

Every extra 1 sqft increases price by ₹3000

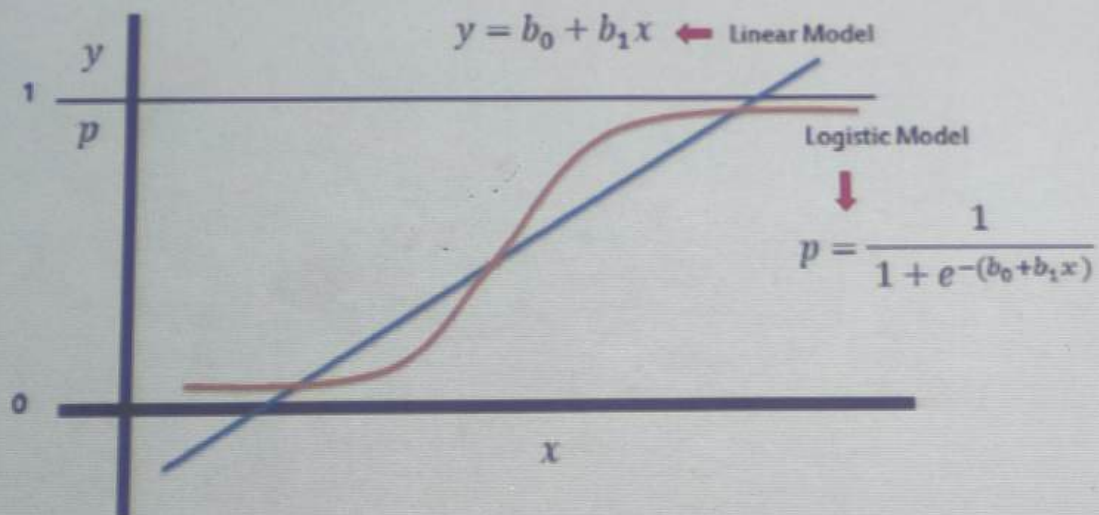
Every extra 1 year of house age decreases price by ₹2000

Base price (when Area = 0 and Age = 0) is ₹50,000

Logistic Regression

- Unlike linear regression which predicts continuous values it predicts the probability that an input belongs to a specific class.
- It is used for binary classification where the output can be one of two possible categories such as Yes/No, True/False or 0/1.
- It uses sigmoid function to convert inputs into a probability value between 0 and 1.
- Logistic Regression is a supervised machine learning algorithm used for classification problems.

Logistic Regression



Applying Interpretability to Logistic Regression

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Ex. You're building a model to predict whether a person likes fast food or not (Yes/No) based on two features: 1. Age 2. Daily Screen Time (in hours)

Person	Age	Screen Time (hrs)	Likes Fast Food?
A	16	6	Yes
B	22	3	Yes
C	30	1	No
D	35	0.5	No
E	19	5	Yes

Applying Interpretability to Logistic Regression

$$\text{Probability of liking fast food} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \cdot \text{Age} + \beta_2 \cdot \text{ScreenTime})}}$$

Let's say the model predicts:

Person X (Age 20, Screen Time 4 hours) $\rightarrow$ Probability = 0.85 $\rightarrow$ YES

Person Y (Age 35, Screen Time 1 hour) $\rightarrow$ Probability = 0.20 $\rightarrow$ NO

You can say:

"More screen time increases the chance of liking fast food"

"Age reduces the likelihood"

These statements are based on the sign and size of the coefficients, which are directly explainable.

Logistic regression doesn't give just a YES or NO — it gives a probability, and you decide the threshold (e.g., 0.5) to classify the result.

Interpretability with Generalized Additive Model (GAM)

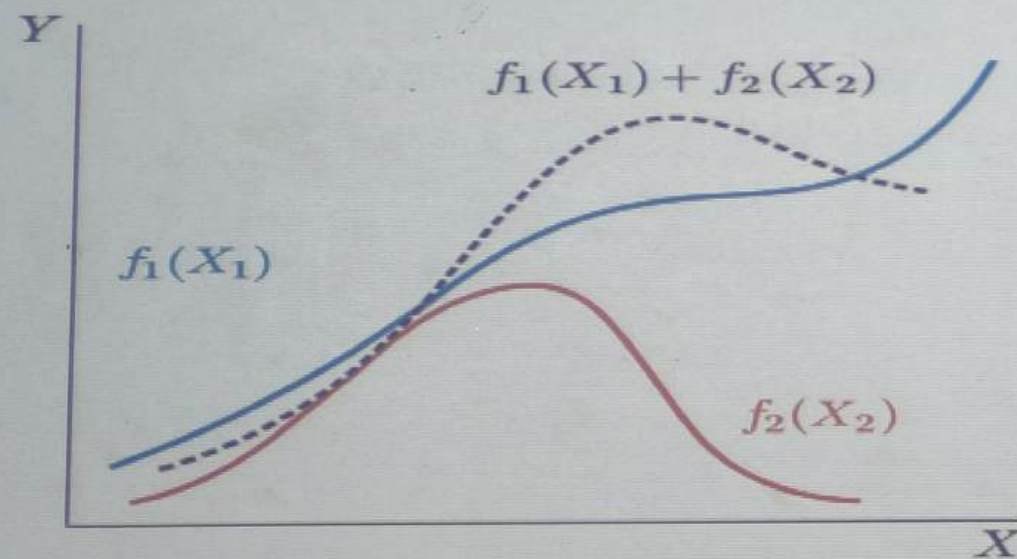
- A Generalized Additive Model (GAM) is an extension of linear models that allows for non-linear relationships between input features and the output, while still maintaining interpretability.
- Unlike linear regression, GAMs model:

$$Y = \beta_0 + f_1(X_1) + f_2(X_2) + \dots + f_n(X_n)$$

- Where each $f_i(X_i)$ is a smooth function (not just a coefficient).
- It keeps the additive structure, so we can isolate and interpret the effect of each feature.

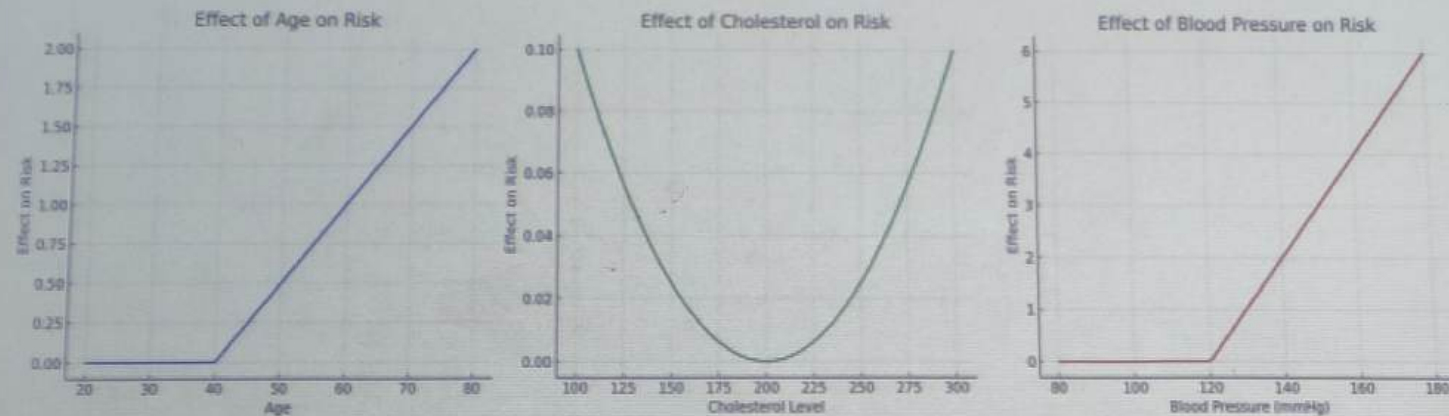
Interpretability with Generalized Additive Model (GAM)

Generalized Additive Models (GAMs)



Interpretability with Generalized Additive Model (GAM)

Ex. Suppose you're predicting medical risk score based on: Age, Blood Pressure, Cholesterol Level



Visualization of GAM: Each feature's effect can be plotted individually like this:

Feature	Shape of $f_i(X_i)$	Interpretation
Age	Gently rising curve	Older people have slightly higher risk
Cholesterol	U-shaped curve	Both low and high levels are risky
Blood Pressure	Flat at first, sharp rise	Risk increases only after a certain value

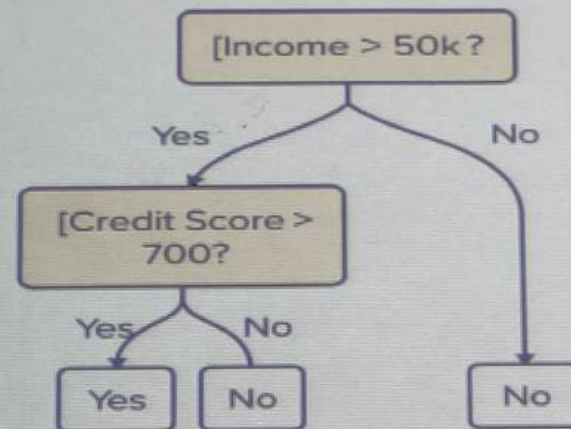
Decision Tree

- A Decision Tree is a supervised learning model used for classification and regression.
- It learns to split the dataset into branches using decision rules based on feature values, forming a tree-like structure.
- At each node, the tree asks a yes/no question about one feature.
- Based on the answer, it splits the data and continues down a branch.
- The process continues until the model predicts an output at the leaf node.

Interpretability with Decision Tree

Ex. Predict/Classify whether a person will buy a car based on: Income, Age, Credit Score

Decision Tree Classifier



For Any Prediction:

"The model predicted YES because:"

Income > ₹50k

Credit Score > 700

→ Therefore, likely to buy a car

Interpretability with Decision Tree

Decision Trees are among the interpretable ML models. Each decision is broken down into simple, logical steps. They are especially useful in education, business rules, and applications where transparency is mandatory.

Why Decision Trees Are Interpretable?

Aspect	Interpretability Value
Visual Structure	You can draw the tree and trace the decision path for any prediction
Clear Rules	Each split is based on simple conditions (e.g., "Age > 30")
Traceability	You can explain exactly why a prediction was made
Human-readable Logic	Mimics human decision-making with IF-THEN rules
Limitation	Explanation
Deep Trees	Too many splits → hard to understand
Not scalable visually	A tree with 100+ nodes is no longer human-readable

Partial Dependence Plot (PDP)

- A Partial Dependence Plot (PDP) is a graph that shows how a single feature (or sometimes two) affects the predictions made by a machine learning model, on average.
- It helps to answer: “How does changing one input affect the model’s output, keeping everything else the same?”
- A Partial Dependence Plot (PDP) is a powerful visualization tool used in machine learning to understand the relationship between a feature (or a pair of features) and the predicted outcome of a trained model — especially complex models like Random Forests or Gradient Boosting.

Partial Dependence Plot (PDP)

Ex. Suppose you built a model to predict if a person will buy health insurance. Your model uses 3 features: Age, Income, Smoking status

Let's say you want to know:

"How does age alone affect the prediction, regardless of income and smoking status?"

How PDP Works (Step-by-step)

1. Take all people in your dataset.
2. For each person, set their age to a fixed value (say, 30), keeping other features unchanged.
3. Predict the outcome (buy insurance or not) using the model.
4. Average all the predictions.
5. Repeat for age = 31, 32, ..., 70.
6. Plot Age vs Average Prediction → PDP

Partial Dependence Plot (PDP)



Partial Dependence Plot (PDP)

Interpretation:

- When age is low (20–30): model predicts low chance of buying insurance.
- After 40: predictions increase — likely to buy insurance.
- After 60: the effect levels off (already high probability).

This shows older people are more likely to buy insurance, based on model predictions.

Point	Explanation
Simple & visual	Easy for non-technical users to understand
Global explanation	Shows the effect of a feature over all data, not just one person
Averages out others	Assumes other features don't change (this can be a limitation sometimes)
Used for any model	Works with Random Forest, XGBoost, Neural Networks, etc.

Surrogate Model

- A Surrogate Model is a simplified approximation (a model that is easier to understand) used to mimic the behavior of a complex, black-box model(Random Forest, XGBoost etc.).
- Trained to replicate the decisions of a complex model, but much simpler and interpretable (e.g., a decision tree or linear model).
- **Common Surrogate Models:**

Surrogate Type	Description
Decision Tree	Most common surrogate; highly interpretable
Linear Model	Simple weighted sum of inputs
Rule-based Model	IF-THEN rules capturing decision patterns

Surrogate Model

Ex. Let's say you build a machine learning model to approve or reject credit card applications.

It uses features like: Age, Monthly Income, Credit Score, Number of Loans, Repayment History

You used a powerful XGBoost model. It gives 95% accuracy — amazing!

But the bank manager asks:

"Why did your model reject Mr. Patel's application?"

XGBoost can't explain itself well. So now..

You Train a Surrogate Model:

1. Take the same dataset (inputs)
2. Use the predictions from XGBoost (not actual labels)
3. Train a Decision Tree to mimic those predictions

If Credit Score > 750 → Approve

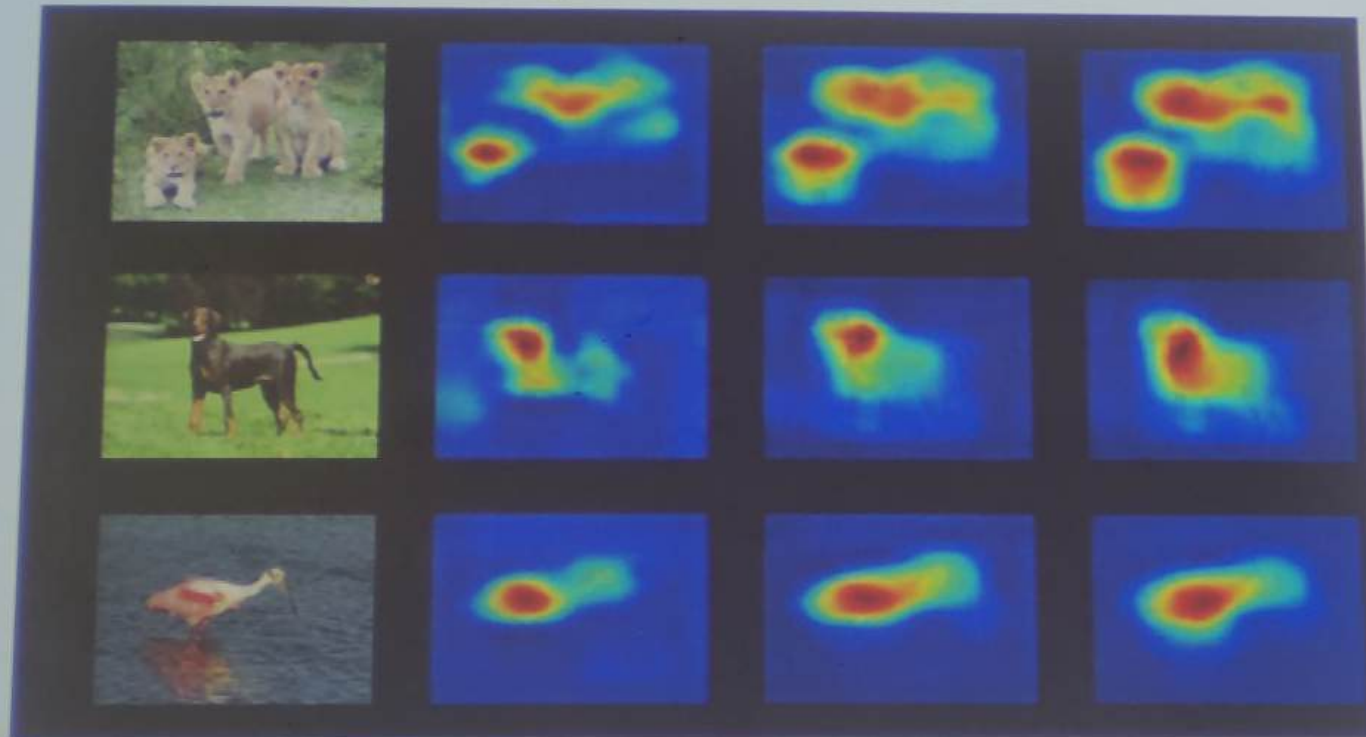
Else If Monthly Income < ₹20,000 → Reject

Else → Approve

Saliency Maps

- Saliency maps are visual representations that highlight the most important regions or features in an image or other data input that contribute to a machine learning model's prediction.
- They essentially act as "heatmaps," where brighter areas indicate higher importance to the model's decision-making process.
- These maps are used in explainable AI (XAI) to help understand why a model made a particular prediction by showing which parts of the input were most influential.
- Primarily used in image or vision processing for Image classification, Object detection, Medical imaging or any model using CNNs (Convolutional Neural Networks)

Saliency Maps



A saliency map may show:

High intensity (red/yellow) around the face and ears of the dog

Low intensity (blue/black) in the background or grass

This tells us the model used the face and ears features to make its decision.

Evaluation of Interpretability in XAI

- Evaluating interpretability means checking how good or useful an explanation is for humans — especially the intended users (e.g., developers, doctors, auditors).
- Not all explanations are equally helpful. Some are too technical, some are too vague, and some might even be misleading.
- So we need ways to evaluate them — both qualitatively and quantitatively.
- XAI is essential-not perfect.

Evaluation Criteria for Interpretability

Criterion	What it checks	Example
Fidelity	How closely the explanation matches the model	A surrogate model mimicking a black-box with high accuracy
Clarity / Simplicity	Is the explanation easy to understand?	A short rule like: "If income < ₹30K → Reject loan"
Stability	Does the explanation change with small input changes?	Same input with slight noise → explanation remains similar
Consistency	Does the method explain similar cases similarly?	Two similar patients → same reason for diagnosis
Actionability	Can users act on the explanation meaningfully?	"Improve credit score" is actionable; SHAP value isn't, alone
Human Trust	Does the explanation increase confidence in model?	A doctor agrees with heatmap over tumor region
Fairness / Ethics	Is the explanation free from bias or discrimination?	Does not blame race or gender